

Sick Days, Snow Days, and the Labor Market Impacts of Caretaking Inequities*

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Abstract

We study how unexpected disruptions impact the labor supply and downstream labor market outcomes of parents in the United States, focusing in particular on events where children cannot attend childcare due to illness or weather-related issues. Using the Current Population Survey and variation at the county-month level in either snowfall or flu-related mortality, we find that mothers bear the entire brunt of these disruptions in terms of missing work. These effects are concentrated among mothers of higher socioeconomic status and have remained stable over the 21st century. Time use data shows that moms reduce their time worked by 12% and increase time spent on childcare by 14% in response to severe weather disruptions. Our findings add to our understanding of the determinants of labor market penalties associated with motherhood and point to the potential usefulness of workplace policies to accommodate childcare problems.

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1 Introduction

Workers are not perfectly substitutable, and there are substantial costs to replacing employees (Jäger et al., 2022). Even temporary absences are costly, with worker absenteeism leading to losses in both wages (Herrmann and Rockoff, 2012), labor productivity (Pauly et al., 2002) and potential reductions in career advancement (Chadi and Goerke, 2018). Along this intensive margin, there may be heterogeneity in external demands on workers' time, which could lead to disparities in who experiences the effects of temporary absences. This paper examines whether unexpected work disruptions impact workers differentially and, if so, who bears the resulting labor market costs.

Using a sample of workers drawn from the Current Population Survey (CPS), with monthly data spanning two decades, we measure the impact of severe flu seasons and high snowfall seasons on individuals' reported ability to work. We gather data on monthly weather patterns from the National Oceanic and Atmospheric Administration (NOAA) and data on flu- and pneumonia-related deaths from the CDC Wide-ranging ONline Data for Epidemiologic Research (WONDER). With these two sources of exogenous variation, we find that unexpected disruptions can impact all workers, but there are distinct gender differences in these effects for parents. Mothers report working fewer hours in months with worse snowfall or flu-related mortality rates compared to fathers. Mothers cite needing to work part-time due to childcare problems at higher rates during these periods of time, but we detect no such effects for fathers and find no such sex-based heterogeneity in impacts for individuals without children. The effects we find are concentrated among mothers of children aged less than 5 and for mothers of relatively high socioeconomic status. We also test whether our estimated impacts have changed between the first two decades of the 21st century and find little evidence of either amplification or attenuation.

Next, we measure the amount of time parents spend working and on childcare using the American Time Use Survey, which we link with our variation on monthly snowfall. Building an indicator for severe within-county weather disruptions measured as a 2 standard deviation increase from the county mean. We find that mothers directly substitute time away from work towards childcare, while fathers make almost no change to how they spend their time. The impact to mothers is substantial: mothers increase their time spent on childcare by 43 minutes (14%) and decrease their time spent working by 42 minutes (12%).

Finally, we use data from the CPS Annual Social and Economic Supplement, which allows

us to determine individuals' snow and illness exposure one year prior, to investigate whether these potential childcare disruptions are associated with measurable decreases in labor supply one year after they happen. We obtain mainly null findings in this analysis, suggesting that mothers may be able to compensate for childcare disruptions with some degree of success. These adjustments themselves are likely to be costly, and the extent to which the need for them factors into maternal preferences for flexible workplace arrangements or inhibits career progression will likely be worthwhile avenues for future research.

This paper's primary contribution comes from making progress in identifying the factors that cause mothers' and fathers' career paths to diverge after parenthood. An extensive literature documents that mothers experience a sizable and lasting drop in earnings after the birth of their first child (Kleven et al., 2019, 2024; Jack et al., 2025). One key factor tied to this trend in mothers' labor supply is childcare availability. Mothers spend more time on caretaking compared to fathers (Cubas et al., 2021; Cowan et al., 2024) and women have a higher demand for flexible work arrangements (Goldin, 2014). Consistent childcare and schooling allows mothers to work and earn more than they do when school and childcare are not available, such as during summer vacation (Price and Wasserman, 2025; Gibbs et al., 2024; Humphries et al., 2024). These gender differences are not solely due to preferences; Buzard et al. (2025) finds that school principals disproportionately call mothers, indicating that external and societal sources play a role in the gendered division of caretaking responsibilities. Our study complements Buzard et al. (2025) well by focusing on external sources that may increase demands for time spent on caretaking.

By using the universal experiences of illness and weather disruptions, we show that mothers take on the bulk of unexpected caretaking demands, leading to short-term reductions in labor supply. While robust childcare and schooling systems may help increase mothers' ability to work, they are not a complete solution to the gap in earnings between mothers and fathers, and additional workplace policies designed to better accommodate unanticipated childcare disruptions may be useful.

The remainder of the paper proceeds as follows. Section 2 describes our sources of data for both dependent and independent variables, along with our empirical strategy for estimating the impacts of potential childcare disruptions on the contemporaneous labor supply of parents. Section 3 describes our headline findings, along with heterogeneity analyses and tests for one-year-out impacts. Section 4 discusses potential directions for future research before concluding.

2 Data and Empirical Strategy

2.1 Current Population Survey Data

Our primary analysis sample comes from the 2000-2019 waves of the Current Population Survey.¹ The CPS has several features that makes it ideally suited for our purposes — its large sample size and time horizon enables us to both detect precise estimates and to study how these estimates have changed over the course of the pre-pandemic 21st century. Moreover, the survey is conducted at a *monthly* level, allowing us to leverage within-year variation in factors such as snowfall and illness propagation, and also contains precise county information for individuals living in sufficiently populous areas,² thereby allowing us precise measurement of exposure to our exogenous variables of interest.

The CPS also contains basic demographic and socioeconomic information that allows us to easily identify parents in the sample and derive basic measures of labor force attachment. Most important, however, is that the CPS contains a vector of questions that makes it ideal for studying the contemporaneous effects of childcare disruptions on labor supply. The CPS asks respondents whether they worked part-time (specifically, fewer than 35 hours) or were absent from work (zero hours) in the previous week, and, if so, why. Potential options include fairly typical reasons, such as slack work conditions, holidays, or the respondent only being able to find part-time work in the first place. However, other reasons are more directly related to our sources of exogenous variation, such as missing work due to weather-related issues, the respondent being sick themselves, or, critically, childcare problems. These responses will form our baseline dependent variables when studying the (very) short-run effects of childcare disruptions on labor force attachment.

We limit our CPS sample to individuals aged 20 to 45 and who are identified as either the household head or the spouse of the household head in the survey and live in a county large enough so as to not be suppressed in the survey. Since we are primarily interested on the impact of childcare disruptions on the contemporaneous labor supply of working parents,

¹We focus on pre-COVID years due to pandemic inducing dramatic fluctuations in both dependent (e.g. part-time work) and independent (e.g. deaths by pneumonia) variables in ways that may be difficult to interpret when pooled with less exceptional years.

²To the extent that more sparsely populated areas have less public transit and longer travel distances, our focus on more populous counties may result in our underestimating the impact of weather-related disruptions on labor supply. While less population density also reduces the transmissibility of disease, our baseline measure of illness will contend with this somewhat by being a per-capita measure.

we additionally limit our sample to individuals who report being employed, and we further limit our sample to individuals surveyed in months when children are likely to be in school (i.e. September through May).³ Table 1 presents summary statistics for the CPS data⁴ in terms of hours worked the previous week as well as the frequency with which respondents report either working part-time or being absent last week for various reasons. As is well-known in the literature, mothers work fewer hours than fathers, and the data suggest that childcare problems are a notable driver of this gap — across the entire sample, mothers are approximately 20 pp more likely to have worked part-time in the previous week than fathers, and slightly over 3 pp (or 15%) of this gap appears to be directly attributable to childcare problems. However, missing work *entirely* due to childcare problems alone appears to be a very rare event, even for mothers of children under 5. Gender differences in part-time usage and hours worked remain for individuals without children as well, though the disparities are considerably smaller in magnitude than those for parents.

2.2 American Time Use Data

We supplement our CPS analysis with a sample drawn from the 2003-2019 waves of the American Time Use Survey (ATUS). Respondents to the ATUS fill out a 24-hour time diary of how they spend their time. Each reported activity is categorized into various types of activities. For our analysis, we focus on two main types of activities: time spent working and time spent on childcare⁵. Time spent on childcare includes both time spent on childcare as a primary activity and as a secondary activity. Mirroring our CPS analysis, we limit our sample to employed respondents. Additionally, we focus on parents by limited to individuals who have their own child in the household. ATUS also provides basic demographic and socioeconomic information which we include in the analysis. Finally, we restrict the main ATUS sample to school-days by only including time diaries from Monday-Friday and during school months October-May when snow is a possibility.

³We also cannot observe whether individuals are unemployed specifically because of childcare-related issues. We will turn to extensive-margin labor supply impacts in Section 3.4.

⁴Descriptive statistics for the subsamples linked to either of our independent variables of interest are indistinguishable from the overall sample.

⁵We consider time spent working as any activity with activity code beginning with “05” and time spent on childcare as any activity with activity code beginning with “03” or time spent on secondary childcare.

2.3 Sources of Exogenous Variation

While the CPS contains a considerable amount of useful information, it does not contain direct information about local weather conditions or the day-to-day health of respondents. We thus supplement our CPS sample with external measures that will be strongly related to the likelihood that households in our CPS sample face childcare disruptions related to weather or illness in a given week. Specifically, we gather data on snowfall and flu-related mortality at the county-year-month level, which we now describe.

Weather data comes from the Global Summary of the Month (GSOM) data files, collected by the National Oceanic and Atmospheric Administration (NOAA). This series aggregates data from weather stations around the world to produce statistics at the station-month level, including variables related to precipitation, temperature, wind, and so on.⁶

We gather data from the universe of stations belonging to the U.S. Cooperative Network, which is the oldest and largest weather network in the United States, having started data collection in 1890. We focus on this particular network due to its complete geographic and temporal coverage of the United States, along with it reliably possessing detailed information pertaining to snowfall that will be key for our research design. We map stations to counties by computing distances of stations from county geographic centroids using longitude and latitude coordinates. For all counties, we average data from all stations within a 50 kilometer radius of the geographic centroid to construct a county-month panel of weather statistics. The key independent variable we gather from this data is the amount of snow (reported in millimeters in the raw data) a given county experienced in the previous month, which we convert to inches for ease of interpretation. Across the United States in the 2000-2019 period, the average snowfall in a given county in a given month is approximately 2 inches, with a standard deviation of around 5 inches.

County-level data on the number of reported flu cases spanning our time period of interest are not reliably available. For exogenous variation related to the probability of illness in a household, we instead use mortality data from the CDC Wide-ranging ONline Data for Epidemiologic Research (WONDER) database. For each county-year-month combination, we take the total number of deaths due to influenza or pneumonia⁷-related reasons before converting these numbers to deaths per 100,000 residents using Census county population

⁶For more documentation, refer to www.ncei.noaa.gov/data/gsom/doc/GSOM_documentation.pdf

⁷Pneumonia being the primary complication from influenza that actually results in fatality.

estimates⁸. While death rates in a county due to influenza are certainly related to the degree of propagation of the disease in the area, they may also relate to the county’s age composition and vaccination rates, since more elderly individuals are much more susceptible to pneumonia than their younger counterparts — to account for this and other county-specific factors, we include county-level fixed effects in our empirical specification.

2.4 Empirical Strategy

Our baseline specification simply studies the impact of increased exposure to snow or illness on labor supply across counties. Specifically, our baseline strategy estimates the following equation:

$$Y_{ictm} = \lambda_c + \tau_t + \gamma_m + \beta \mathbf{X}_i + \delta T_{ctm} + \varepsilon_{ictm}, \quad (1)$$

where Y_{ictm} is the outcome of interest (e.g., hours worked the previous week or working part-time last week due to childcare problems) for individual i in the CPS, measured in county c in year t and month m . We include fixed effects for county, year, and month to account for geographic, annual, and seasonal trends in the dependent and treatment variables. We additionally include vectors of demographic characteristics $\mathbf{X}_i$ to account for individual factors that may relate to the dependent or independent variable, including controls for race, marital status, education, and a cubic polynomial in age.

The parameter of interest is thus δ , which captures the impact of a unit increase in the independent variable T_{ctm} (namely, inches of snow or flu deaths per capita) in the relevant county-year-month. Identifying variation comes from county-level differences in snowfall or illness relative to trend, which may happen within counties across time, across counties within time, and combinations of the two. The identifying assumption is that, after accounting for geographic and temporal trends, changes in snowfall or mortality rates at the county level are unrelated to any other factors that may impact short-run labor supply decisions. While this baseline assumption is strong, the features of our data allow us to conduct myriad face validity and placebo tests along with robustness exercises that focus on more heavily restricted sources of identifying variation.

⁸We choose this scaling for ease of interpretation, as an additional death per 100,000 residents is roughly equivalent to a one-standard-deviation increase, with the mean rate per 100k being 1.67. The CDC reports when a county has zero deaths in a month but suppresses the exact count if it is between 0 and 10 — we code such cases as missing for lack of a better alternative.

We also include a within-county specification for our ATUS analysis, building indicators for monthly snowfall more than one and two standard deviations above the county mean, and estimate 1 replacing T_{ctm} with each of these indicators. This measure allows us to estimate the causal effect of experiencing a severe event for the area the individual lives.

3 Results

3.1 Contemporaneous Effects

Figures 1 and 2 present our main findings for the contemporaneous effects of increased snow or illness on the labor supply of respondents in the CPS. All numbers in the table are separate estimates of δ from Equation 1 and their standard errors. Binary outcome variables, such as part-time status, are scaled to be either 0 or 100, so that the estimates are interpretable as percentage point changes in the outcome of interest.

Several features of the results warrant discussion. The first is that, among parents (the final four columns in any of the subfigures), the impacts of increased exposure to either snow or illness on labor supply are almost exclusively loaded onto mothers. This is especially true for parents of children less than 5 years old. Increases in monthly snowfall or flu-related mortality are associated with reductions in hours worked the previous week throughout the CPS sample, but the effects are the most significant (both statistically and substantively) for the mothers of young children — an additional inch of snowfall in the county-month is associated with 0.026 (Figure 1a) fewer hours worked the previous week for such mothers, while the corresponding effect for an additional flu-related death per 100,000 residents in the county-month is 0.1 (Figure 2a).

We perform the following calculation to put the magnitude of our results into context, noting the exogenous variables are measured at the monthly level while the outcome variables ask about labor supply in the preceding week. Taking the snow coefficient for the mothers of young children as an example, consider the case of a county receiving a foot of snow in a given month (around two standard deviations above the mean). The impact of hours worked for the month may be approximated as $0.026 \times (12 \text{ inches}) \times (4 \text{ weeks}) = 1.25$ hours, which on a basis of an average of 32.3 weekly hours worked for the mothers of young children translates to around a 1 percent reduction in hours worked each month. Considering a similar 2σ scenario for flu mortality rates yields a remarkably similar figure of 1.4 hours, though we

acknowledge that the latter estimate is noisily estimated. The magnitudes of our estimates may also be usefully interpreted through the baseline gap in hours worked between mothers and fathers: with mothers of young children working on average 10 fewer hours per week than fathers, or 40 fewer hours per month, the aforementioned numbers would constitute a roughly 3 percent increase in the parental labor supply gap.⁹

While worse weather or more widespread illness could depress labor supply through a number of channels, our results also directly point to childcare disruptions as being a key driver in the heterogeneity of effects between mothers and fathers. When looking at the reasons given for working part-time in the previous week, the rate at which fathers report missing work due to childcare problems (Figures 1d and 2d) is a precisely estimated zero, regardless of treatment or whether the father has young children in the household. This is emphatically not the case for mothers, who consistently cite childcare problems as forcing them to work part-time — for instance, an additional inch of snow (flu death per 100k) increases the likelihood that mothers of young children work part time due to childcare problems by 0.038 (0.205) percentage points, which on the base rate of approximately 5 percent constitutes increases of 0.76 (4.1) percent in any given week.

One limitation of our measure of snowfall is that snow is not distributed randomly across the country—as such, a county that receives an additional inch of snowfall is more likely to be a county that possesses the necessary equipment or infrastructure to minimize the disruption caused by it. As an alternate experiment, we construct an indicator variable that captures whether a county experienced a *within-county* two standard deviation or higher monthly snowfall total. While not as directly interpretable as an inch of snow, this measure captures a weather event that is more *unusual* for the location, which may range from minimal snowfall in Southern locations to multiple feet of snow in Northern ones. The results of this exercise, shown in Figure A.II, exhibit a similar qualitative pattern of our baseline findings but with sensibly much larger magnitudes — for the mothers of young children, a within-county two standard deviation snowfall raises the probability of working part-time due to childcare problems by over 0.5 pp (10%) and results in 0.4 fewer hours worked per week.

⁹Nonlinearities in the effects of snow or flu exposure may also drive our estimates down as well as the fact that the true impacts of more snow or illness on labor supply are likely bimodal, with the majority of parents being unaffected and a subset losing more substantial work time. We observe similar gendered heterogeneity patterns when looking at the impacts of snowfall on an indicator variable for having missed eight hours or more in the previous week (Figures 1b and 2b), but the associations between this outcome and increased illness are considerably less precise.

Mothers in general also appear substantially more likely to miss 8 hours or more of work in the previous week in response to this sort of snowfall shock.

Our results also pass several face validity and placebo tests. For instance, we sensibly find that increased snowfall makes all CPS respondents more likely to miss work due to weather-related problems with limited heterogeneity by sex (Figure 1c); the effects of increased flu mortality on missing work due to one’s own illness follow a qualitatively similar pattern (Figure 2c). Moreover, the effects of our treatments on the labor supply of CPS respondents *without children* show either negligible heterogeneity by sex or larger effects for men, such that an alternative empirical approach that estimated impacts on mothers by “differencing out” the effects for men and women without children would either maintain or increase our baseline point estimates.¹⁰

We consider additional placebo tests in Table A.I, where we estimate Equation 1 for the outcome variables of *usual* (as opposed to actual) weekly hours worked and an indicator for the respondent being *absent* from work (that is, missing work entirely) in the previous week due to childcare problems. Significant associations between our treatment variables and usual weekly hours would raise concerns that our baseline estimates are in fact either detecting a spurious relationship or are contaminated by omitted variable bias, while the near-zero frequency of absence from work due to childcare problems observed in Table 1 suggests that we should anticipate null effects for this outcome variable as well. Reassuringly, we obtain nearly exclusively precise null estimates for both placebo outcome variables. An exception to this is the impact of flu exposure to usual hours worked, which exhibits a negative and noisy association. This finding, along with the noisier baseline estimates of flu exposure on maternal labor supply, guides our decision to direct our focus to the weather-related results going forward. We also consider more stringent specifications in Table A.II, where we include either year-by-month or state-by-year fixed effects in our estimating equation, thereby substantially limiting our identifying variation to within-month-cross-county or cross-month-within-county differences in treatment. Even with this limited source of identifying variation, we find quite similar results to our baseline estimates. We investigate

¹⁰We also note that we would not necessarily expect the impacts of increased exposure to snow or illness on hours worked for individuals without children to be zero, since they, too, may be influenced by weather or sickness-related disruptions. Indeed, in some cases the estimated magnitudes are *larger* for individuals without children, potentially pointing to their labor supply being slightly more elastic in response to inconveniences than parents. Flu exposure in particular may increase demands for informal care for elderly parents — while these obligations, too, tend to be gendered in nature (Mommaerts, 2025), there is less reason to expect that this heterogeneity would be stronger for parents than childless individuals.

potential non-linearities in impacts by producing binned scatter plots of residualized outcome and independent variables (Figure A.I) but find little evidence of non-linear effects.

3.2 Heterogeneity

We now investigate heterogeneity in the impacts of potential childcare disruptions on labor supply over demographic and socioeconomic characteristics. The results of these analyses are presented in Figure 3. For all analyses, we focus on the mothers of young children, as this was the subgroup for whom we identified the largest impacts on Section 3.1. We report impacts on hours worked and whether the respondent worked part-time due to childcare issues — additionally, to improve visual interpretability we flip the sign on the estimated impacts for hours worked, so the estimates may be interpreted as hours *lost* due to increased exposure to our exogenous variables.

The main thematic result we obtain from this exercise is that the contemporaneous impacts of increased snow or illness on labor supply appear to be larger for individuals that are of higher socioeconomic status. We typically find larger impacts on labor supply for white mothers than Black mothers, for single mothers than married mothers, and for college-educated women than non-college-educated women. These findings are consistent with several possible explanations, such as the opportunity cost of taking time off work being lower when there are multiple earners in the household or Black and single women relying more heavily on informal childcare than their white or married counterparts (Anstreicher and Venator, 2025). It is important to distinguish this story, however, from married women taking time off work more specifically because they earn *less* than their spouses — when we focus on married women who had *higher* income from wages and salary in the previous year¹¹, we continue to obtain positive and statistically significant estimates, including some that are larger than for mothers who are *not* the household’s primary earner¹².

Taken together, our results suggest that mothers bear the brunt of childcare disruptions in terms of their labor supply adjustments and that higher-SES mothers are more heavily impacted than lower-SES mothers. As a final heterogeneity exercise, we test whether our impacts have changed over the course of the 21st century by separately estimating impacts in the 2000-2009 and the 2010-2019 window. We do not observe any clear or consistent

¹¹Available from the Annual Social and Economic Supplement in the CPS, administered in March.

¹²Our focus on increasingly small subsamples of the CPS results in statistical noise such that we are often unable to reject the equivalence of estimated coefficients with a strong degree of confidence.

patterns of heterogeneity across these two time periods, which provides some evidence that the impacts we find are persistent and unlikely to go away on their own accord.

3.3 American Time Use Survey

Figure 4 presents our main findings for the impact of severe within-county weather on parental time use. Each reported change is the estimate of δ from separate within-county versions of Equation 1, using indicators for severe weather. Estimates of each δ and their standard deviations are also reported in A.III along with estimates for validation comparisons described below.

Notably, mothers spend more time on childcare and less time working than fathers on average, even before severe disruptions. Weather-related disruptions exacerbate this disparity. In response to a 2 standard deviation increase in monthly snowfall from the county mean, mothers report significantly increasing their time spent on childcare by 43 minutes, a 14% increase from the baseline mean. Mothers significantly decrease their time spent working by 42 minutes, a 12% decrease from the baseline mean. This almost exact substitution of time points to mothers taking on the full burden of these child-related disruptions. Fathers, by comparison, slightly increase their time spent working and decrease their time spent on childcare, although neither of these findings are significant.

These differences are driven by disruptions to days when children are typically in childcare, and only exist for individuals with children. Figure A.III shows no significant difference between mothers and fathers of the impact of these severe disruptions on weekends. Additionally, there are no gender differences across individuals who do not have children, meaning that there are no underlying gender differences in response to severe weather without the presence of children.

3.4 Downstream Labor Market Impacts

Our results above demonstrate that increased exposure to adverse weather or illness generates childcare disruptions that negatively impact the labor supply of mothers. Are these contemporaneous effects associated with longer-run impacts in labor supply and employment? To probe this question, we evaluate one-year-out impacts of increased exposure to the same sources of exogenous variation using CPS Annual Social and Economic Supplement (ASEC) data. This supplement, conducted yearly in March, asks additional questions that enable us to determine previous-year exposure to snow or illness with some confidence and

so estimate the downstream labor market impacts of potential childcare disruptions.

In particular, the ASEC asks respondents whether the respondent moved across counties in the previous year. We limit our sample as before to household heads and their spouses aged 20 to 45 and then restrict our sample further to individuals who made no such move so that their current county of residence identifies their county from one year ago. We then regress labor market outcomes that are less likely to immediately response to childcare disruptions — including wage income in the previous year, indicators for employment and labor force non-participation, and usual hours worked weekly — on the county’s exposure to snow or illness in the *previous year*, which we define as either total snowfall, in inches, in the previous calendar year’s winter months¹³, or average county-level flu and pneumonia-related deaths per 100,000 residents in the same time window. Conceptually, these tests are intended to focus less on the immediate impacts of childcare disruption and more on whether potential childcare disruptions in the past have measurable impacts on labor market outcomes in the present.

The results of this exercise are presented in Table 2, with the headline finding being that we do not detect any statistically significant impacts of potential childcare disruptions on labor market outcomes one year after they happen. Virtually all estimated impacts for previous-year snowfall are either statistically insignificant or do not exhibit meaningful heterogeneity by sex. While some heterogeneity in impacts between mothers and fathers appears to manifest when looking at the impacts of previous-year exposure to illness, with mothers showing slight decreases in wage income, employment, and usual hours worked, these patterns are both noisily estimated and frequently do not differ qualitatively from patterns found for individuals without children¹⁴

We discourage interpreting these findings as evidence that the contemporaneous impacts we detected earlier are unimportant. The null effects we find here are consistent with mothers being able to compensate for lost work in the face of childcare disruptions to a degree, such

¹³Defined as December, January or February

¹⁴We extend this analysis by conducting a similar specification using the American Community Survey (ACS). The ACS has the advantage of being significantly larger than the CPS ASEC but has downsides in exact month of survey not being available in the public-use version, which introduces some imprecision in event time. Estimates from the ACS thus have the interpretation of being pooled effects one to two years after exposure to increased illness or snow. The results, available in Table A.V, nonetheless are qualitatively quite similar to the ASEC estimates, featuring statistically and substantively insignificant results that are inconsistent with large effects in either direction.

as by taking paid time off or through working additional hours in subsequent weeks. This does not take away from the fact that the burden of covering work time lost due to childcare disruptions appears to primarily fall on mothers, and these adjustments themselves may be costly and inconvenient. Moreover, the consistent threat of childcare disruptions may drive mothers into pre-emptively selecting into more flexible work arrangements before these childcare disruptions occur, which in turn may inhibit their career progression (Goldin, 2014) and attenuate our estimates. Detecting such a long-run association would clearly require a different sample than the one we use here.

4 Discussion

This paper adds to the ongoing effort to unpack the determinants of the labor market penalty that mothers experience by focusing on the impacts of childcare disruptions that virtually all parents face at one point or another: sick days and snow days. By combining data from the Current Population Survey with data on snowfall and flu-related mortality, all at the county-month level, we document that increased exposure to potential childcare disruptions results in contemporaneous decreases in labor supply that are strongly concentrated among mothers, especially high-socioeconomic-status mothers with young children. While we do not detect that these immediate impacts result in adverse labor market outcomes in the longer run, we also find that they have been relatively stable over the course of the 21st century. In addition, maternal expectations about future childcare disruptions may contribute towards their selection into more flexible or part-time employment, which is a question for future research along with how our findings may manifest in longer-term career advancement outcomes.

One interesting feature of our results is that we consistently estimate stronger impacts of snowfall and illness on the labor supply of the mothers of young children specifically. In the case of illness, this may be explained by increased transmissibility of disease among younger children, but the stronger impacts for increased snow exposure may be surprising to the extent that snow days primarily impact older children already enrolled in public school systems. We interpret these results as potentially indicating increased sensitivity of private childcare centers in response to weather events or increased difficulty in finding alternative or informal care arrangements for younger children than for older children. At the same time, we note that being the parent of a young child in our sample does not preclude one

from also having older children in public school, and being a parent in general also includes individuals with older children who require substantially less supervision. We investigate these possibilities further in Table A.III, which interacts our treatment variables by the number of children in the household aged 0 to 5, 6 to 11, and 12 to 18 for mothers in the CPS, and continue to find that effects are most strongly loaded on mothers with younger children. However, further research into the interactions of child age and potential labor supply disruptors may be useful.

While reliable childcare and schooling encourage maternal labor force participation, these two systems alone are not sufficient to close the existing gender gap in the labor market. Unexpected caretaking responsibilities, which are not covered by these systems, fall disproportionately on mothers. While there are various policies that may aim to address this gender gap, we also highlight the socioeconomic gap in mothers' caretaking in response to illness and weather-related events. This additional gap suggests that access to resources affects parents' ability to take time away from work when a child needs extra care. Related policies, such as paid sick time or flexible work arrangements, would need to be evaluated along both dimensions if intended to increase equity in the labor market.

References

- Anstreicher, G. and Venator, J. (2025). To grandmother’s house we go: Informal childcare and female labor mobility.
- Buzard, K., Gee, L. K., and Stoddard, O. (2025). Who you gonna call? Gender inequality in external demands for parental involvement. *The Quarterly Journal of Economics*, 140:2805–2849.
- Centers for Disease Control and Prevention (2025). CDC WONDER.
- Chadi, A. and Goerke, L. (2018). Missing at work – sickness-related absence and subsequent career events. *Journal of Economic Behavior & Organization*, 153:153–176.
- Cowan, B., Jones, T. R., and Swigert, J. (2024). Parental and student time use around the academic year. *Journal of Economic Behavior and Organization*, 224:66–110.
- Cubas, G., Juhn, C., and Silos, P. (2021). Work-care balance over the day and the gender wage gap. *AEA Papers and Proceedings*, 111:145–153.
- Flood, S., King, M., Rodgers, R., Ruggles, S., Warren, J. R., Backman, D., Breton, E., Cooper, G., Drew, J. A. R., Richards, S., and Riper, D. V. (2025). Integrated public use microdata series, current population survey: Version 13.0 [dataset]. *IPUMS*.
- Gibbs, C., Wikle, J., and Wilson, R. (2024). A matter of time? measuring effects of public schooling expansions on families’ constraint. *SSRN Electronic Journal*.
- Goldin, C. (2014). A grand gender convergence: Its last chapter. *American Economic Review*, 104:1091–1119.
- Herrmann, M. A. and Rockoff, J. E. (2012). Worker absence and productivity: Evidence from teaching. *Journal of Labor Economics*, 30:749–782.
- Humphries, J. E., Neilson, C., Ye, X., and Zimmerman, S. D. (2024). Parents’ earnings and the returns to universal pre-kindergarten.
- Jack, R., Tannenbaum, D., and Timpe, B. (2025). The parenthood gap: Firms and earnings inequality after kids. *Upjohn Institute Working Papers*.

- Jäger, S., Heining, J., and Lazarus, N. (2022). How substitutable are workers? Evidence from worker deaths *. *National Bureau of Economic Research, No. w30629*.
- Kleven, H., Landais, C., and Leite-Mariante, G. (2024). The child penalty atlas *. *Review of Economic Studies*.
- Kleven, H., Landais, C., and Søgaaard, J. E. (2019). Children and gender inequality: Evidence from denmark. *American Economic Journal: Applied Economics*, 11:181–209.
- Lawrimore, J. H., Ray, R., Applequist, S., Korzeniewski, B., and Menne, M. J. (2016). Global summary of the month (GSOM), version 1. *NOAA National Centers for Environmental Information*.
- Mommaerts, C. (2025). Long-term care insurance and the family. *Journal of Political Economy*, 133(1):1–52.
- Pauly, M. V., Nicholson, S., Xu, J., Polsky, D., Danzon, P. M., Murray, J. F., and Berger, M. L. (2002). A general model of the impact of absenteeism on employers and employees. *Health Economics*, 11:221–231.
- Price, B. and Wasserman, M. (2025). The summer drop in female employment. *Review of Economics and Statistics*.

Table 1: Summary Statistics of CPS Sample

SAMPLE	All	Men, No Kids	Women, No Kids	Fathers, All	Mothers, All	Fathers, Kid < 5yo	Mothers, Kid < 5yo
VARIABLE							
Age	34.7 (6.7)	32.5 (6.8)	32.0 (7.0)	36.8 (5.7)	36.2 (6.0)	34.3 (5.5)	32.6 (5.6)
White (%)	76.1 (42.7)	76.3 (42.5)	74.5 (43.6)	79.7 (40.2)	73.2 (44.3)	79.4 (40.4)	73.5 (44.1)
Black (%)	13.2 (33.9)	12.6 (33.1)	14.2 (34.9)	9.8 (29.8)	16.6 (37.2)	9.3 (29.1)	15.7 (36.4)
College Degree (%)	41.8 (49.3)	43.6 (49.6)	55.0 (49.7)	36.8 (48.2)	37.0 (48.3)	41.4 (49.2)	44.2 (49.7)
Hours worked last week	39.0 (13.4)	40.8 (13.1)	38.1 (12.9)	42.3 (12.6)	34.6 (13.5)	42.2 (12.7)	32.7 (14.4)
Part-time last week (%)	18.7 (39.0)	14.4 (35.1)	20.4 (40.3)	10.7 (30.9)	29.3 (45.5)	11.3 (31.6)	32.2 (46.7)
<i>Due to weather</i>	0.3 (5.8)	0.4 (6.0)	0.2 (4.6)	0.5 (6.9)	0.2 (5.0)	0.5 (7.0)	0.2 (4.8)
<i>Due to illness</i>	1.4 (11.8)	1.2 (10.8)	1.7 (13.0)	1.1 (10.4)	1.8 (13.2)	1.1 (10.4)	1.7 (13.0)
<i>Due to childcare problems</i>	1.1 (10.6)	0.0 (1.9)	0.2 (4.3)	0.3 (5.2)	3.5 (18.5)	0.3 (5.9)	5.4 (22.5)
Absent from work last week (%)	2.8 (16.5)	2.3 (14.9)	2.7 (16.2)	2.1 (14.3)	4.1 (19.8)	2.2 (14.6)	6.3 (24.3)
<i>Due to weather</i>	0.1 (2.7)	0.1 (2.8)	0.0 (1.5)	0.1 (3.6)	0.0 (2.1)	0.1 (3.4)	0.0 (2.1)
<i>Due to illness</i>	0.6 (7.5)	0.4 (6.6)	0.6 (7.4)	0.4 (6.6)	0.8 (8.8)	0.4 (6.3)	0.7 (8.4)
<i>Due to childcare problems</i>	0.0 (2.0)	0.0 (0.7)	0.0 (1.0)	0.0 (1.3)	0.1 (3.3)	0.0 (1.7)	0.2 (4.4)
<i>N</i>	2102730	489628	390320	595415	627367	294370	239240

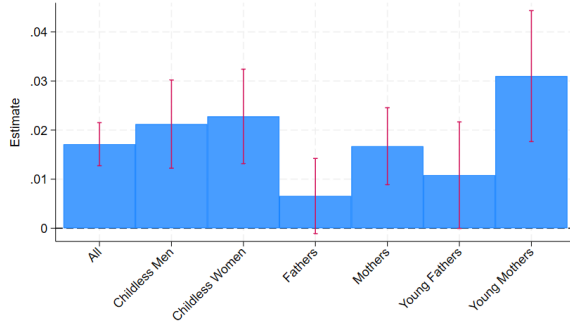
Notes: Means for each variable presented above with standard deviations included in parentheses. Data from 2000-2019 waves of the Current Population Survey. Data restricted to employed household heads and spouses of household heads age 20-45; see text for details. Absence from work defined as working zero hours in previous week. Part-time work defined as working positive but fewer than 35 hours in previous week. Mothers and fathers with kid under 5 are defined as fathers or mothers with at least one own child age less than 5 present in the household.

Table 2: Downstream Labor Supply Impacts

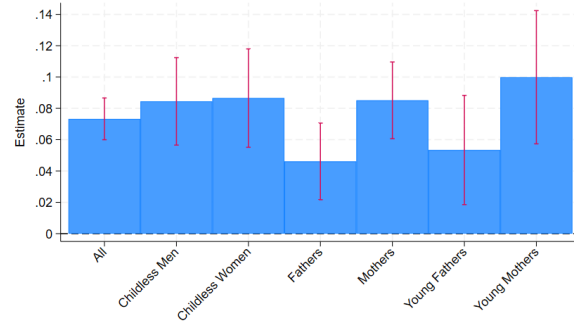
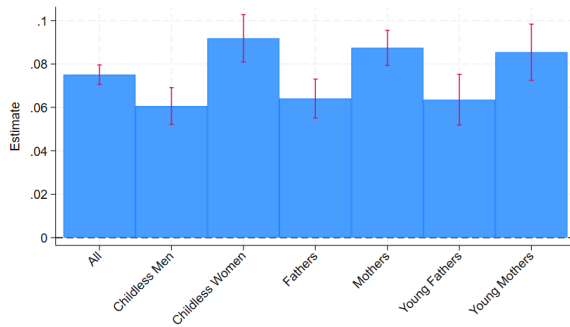
Panel A: Effects of Increased Snow							
SAMPLE	All	Men, No Kids	Women, No Kids	Fathers, All	Mothers, All	Fathers, Kid < 5yo	Mothers, Kid < 5yo
VARIABLE							
Wage Income	123.0 (68.1)	71.9 (168.0)	34.4 (153.0)	167.0 (139.0)	177.0 (83.1)	238.0 (195.0)	-23.5 (127.0)
Employed	0.085 (0.079)	-0.093 (0.188)	0.083 (0.215)	0.122 (0.104)	0.148 (0.136)	0.109 (0.151)	-0.212 (0.216)
Not in Labor Force	-0.017 (0.071)	0.004 (0.150)	0.079 (0.196)	0.019 (0.075)	-0.128 (0.130)	0.032 (0.104)	0.298 (0.210)
Usual Hours Worked	0.116 (0.035)	-0.043 (0.087)	0.097 (0.092)	0.122 (0.054)	0.226 (0.052)	0.130 (0.079)	0.135 (0.082)
Panel B: Effects of Increased Illness							
VARIABLE							
Wage Income	-163.0 (105.0)	-153.0 (262.0)	-103.0 (244.0)	-239.0 (202.0)	-200.0 (127.0)	-154.0 (298.0)	-429.0 (190.0)
Employed	-0.052 (0.135)	-0.130 (0.307)	-0.425 (0.384)	0.226 (0.144)	-0.180 (0.231)	-0.009 (0.224)	-0.351 (0.375)
Not in Labor Force	0.231 (0.125)	0.486 (0.250)	0.726 (0.344)	-0.074 (0.118)	0.189 (0.225)	0.093 (0.180)	0.368 (0.372)
Usual Hours Worked	0.039 (0.061)	0.177 (0.148)	0.165 (0.164)	0.153 (0.089)	-0.166 (0.090)	0.093 (0.142)	-0.187 (0.141)

Notes: Table presents estimates of δ from Equation 1 for indicated dependent variables and samples. Robust standard errors in parentheses. Data from 2000-2019 waves of the Current Population Survey ASEC, restricted to employed household heads and spouses of household heads age 20-45 who did not move across counties in the previous calendar year. Snow measured in feet of accumulation in the September-May of the previous calendar year in the relevant county. Illness measured in average deaths per 100,000 residents due to flu or pneumonia-related reasons in relevant county in the September-May of the previous calendar year. See text for details. Mothers and fathers with kid under 5 defined as fathers or mothers with at least one own child less than 5 present in the household.

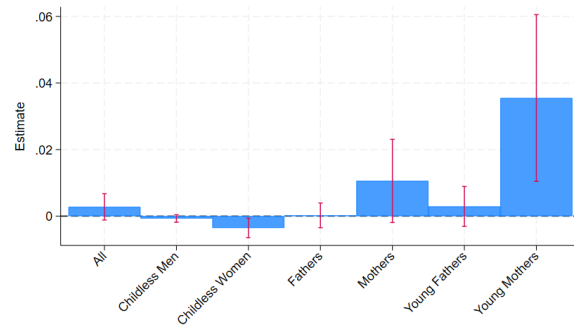
Figure 1: Baseline Results — Effects of Increased Snow



(a) Hours Missed

(b) Missed ≥ 8 Hours

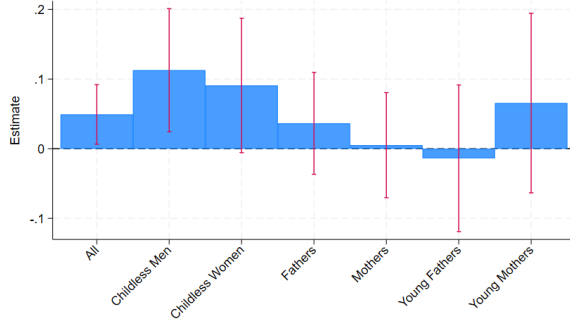
(c) Part-time for Weather



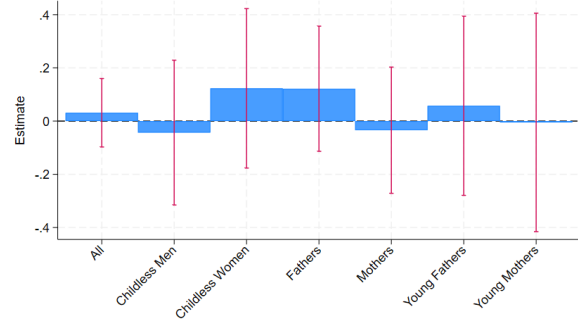
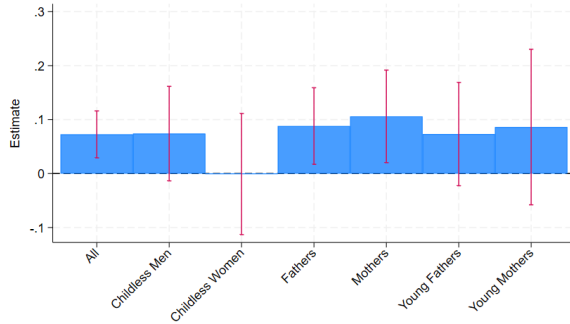
(d) Part-time for Childcare Problems

Notes: Young fathers/mothers defined as fathers or mothers with at least one own child age less than 5 present in the household. Figures presents estimates of δ from Equation 1 for indicated dependent variables and samples. 95% confidence intervals in red. Data from 2000-2019 waves of the Current Population Survey, restricted to employed household heads and spouses of household heads age 20-45. Part-time work defined as working positive but fewer than 35 hours in previous week. Snow measured in inches of accumulation in relevant county-month. See text for details.

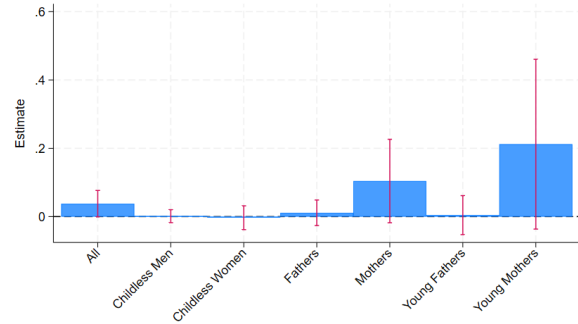
Figure 2: Baseline Results — Effects of Increased Illness



(a) Hours Missed

(b) Missed ≥ 8 Hours

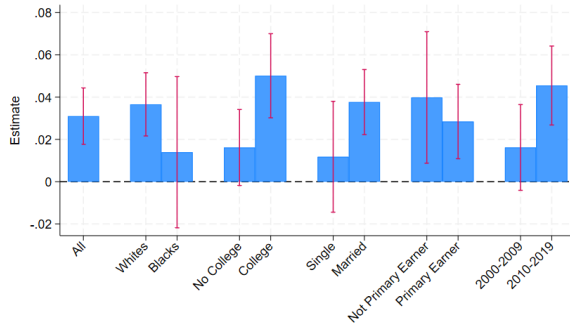
(c) Part-time for Own Illness



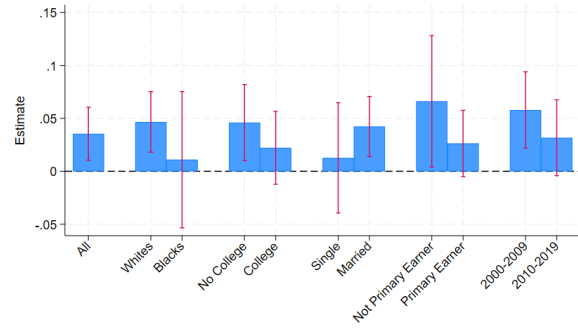
(d) Part-time for Childcare Problems

Notes: Young fathers/mothers defined as fathers or mothers with at least one own child age less than 5 present in the household. Figures presents estimates of δ from Equation 1 for indicated dependent variables and samples. 95% confidence intervals in red. Data from 2000-2019 waves of the Current Population Survey, restricted to employed household heads and spouses of household heads age 20-45. Part-time work defined as working positive but fewer than 35 hours in previous week. Illness measured in deaths per 100,000 residents due to flu or pneumonia-related reasons in relevant county-month. See text for details.

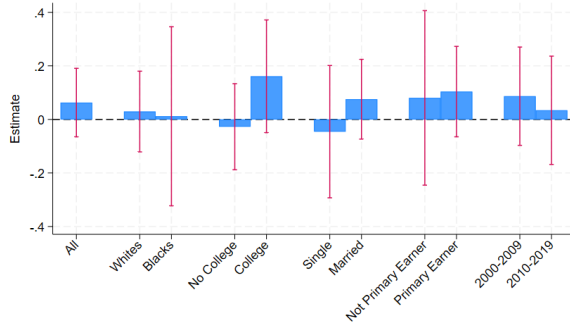
Figure 3: Heterogeneity Analyses



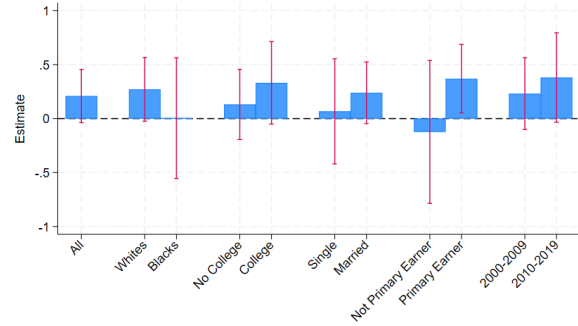
(a) Snow, Hours Missed



(b) Snow, Part-Time for Childcare



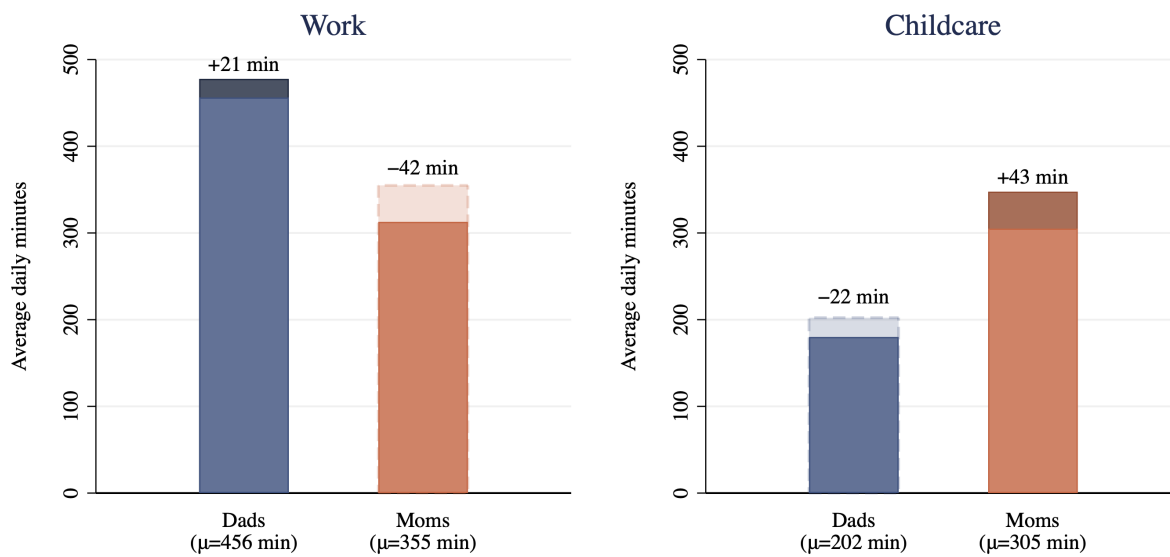
(c) Illness, Hours Missed



(d) Illness, Part-Time for Childcare

Notes: Figure presents estimates of δ for indicated subgroups of mothers of children age less than 5 in the CPS. Red lines indicate 95% confidence intervals of estimates. Data from 2000-2019 waves of the Current Population Survey, restricted to employed household heads and spouses of household heads age 20-45. Part-time work defined as working positive but fewer than 35 hours in previous week. Primary earner status defined for married mothers as having the highest wage/salary income among household members in the previous year. Snow measured in inches of accumulation in relevant county-month. Illness measured in deaths per 100,000 residents due to flu or pneumonia-related reasons in relevant county-month. See text for details.

Figure 4: Parental Time Use



Notes: Figure presents estimates of parental time spent on work and childcare overall and the change in response to a within-county 2 standard deviation increase in monthly snowfall. Data from 2003-2019 waves of the American Time Use Survey, restricted to employed parents with own child in household. Childcare minutes include all time spent on both primary childcare (activity category 3) and secondary childcare (*SCC_ALL*); work minutes include all work-related activities (activity category 5); see text for details.

A Supplementary Tables and Figures

Table A.I: Placebo Tests: Impacts on Usual Hours Worked and Work Absence

Panel A: Effects of Increased Snow							
SAMPLE	All	Men, No Kids	Women, No Kids	Fathers, All	Mothers, All	Fathers, Kid < 5yo	Mothers, Kid < 5yo
VARIABLE							
Usual hours worked	−0.003	−0.002	−0.004	00	−0.001	−0.004	−0.010
weekly	(0.002)	(0.004)	(0.004)	(0.003)	(0.003)	(0.004)	(0.006)
Absent due to	0.000	0.000	0.000	0.000	0.000	0.000	0.001
childcare	(0.000)	(0.000)	(0.000)	(0.000)	(0.001)	(0.001)	(0.002)
Panel B: Effects of Increased Illness							
VARIABLE							
Usual hours worked,	−0.028	−0.052	−0.054	0.004	−0.035	0.030	−0.090
weekly	(0.018)	(0.037)	(0.041)	(0.030)	(0.034)	(0.043)	(0.059)
Absent due to	0.003	0.000	−0.006	0.003	0.008	0.001	−0.010
childcare	(0.004)	(0.001)	(0.003)	(0.003)	(0.012)	(0.004)	(0.023)

Notes: Mothers and fathers with kid under 5 defined as fathers or mothers with at least one own child age less than 5 present in the household. Table presents estimates of δ from Equation 1 for indicated dependent variables and samples. Robust standard errors in parentheses. Data from 2000-2019 waves of the Current Population Survey, restricted to employed household heads and spouses of household heads age 20-45. Absence from work defined as working exactly zero hours in the previous week. Snow measured in inches of accumulation in relevant county-month. Illness measured in deaths per 100,000 residents due to flu or pneumonia-related reasons in relevant county-month. See text for details.

Table A.II: Robustness to Alternate Fixed Effects Structures

Panel A: Effects of Increased Snow									
SAMPLE	Fathers, All	Mothers, All	Fathers, Kid < 5	Mothers, Kid < 5	Fathers, All	Mothers, All	Fathers, Kid < 5	Mothers, Kid < 5	Mothers, Kid < 5
VARIABLE									
Hours worked last week	-0.001 (0.004)	-0.013 (0.004)	-0.004 (0.006)	-0.029 (0.007)	-0.008 (0.004)	-0.014 (0.004)	-0.014 (0.006)	-0.030 (0.007)	
Part-time due to childcare	-0.001 (0.002)	0.013 (0.007)	0.004 (0.003)	0.045 (0.013)	0.000 (0.002)	0.007 (0.007)	0.003 (0.003)	0.028 (0.013)	
Panel B: Effects of Increased Illness									
VARIABLE									
Hours worked last week	-0.028 (0.041)	0.014 (0.041)	0.028 (0.059)	-0.057 (0.071)	-0.082 (0.039)	0.004 (0.041)	-0.035 (0.057)	-0.046 (0.071)	
Part-time due to childcare	0.010 (0.019)	0.079 (0.067)	0.007 (0.029)	0.150 (0.136)	0.014 (0.020)	0.061 (0.068)	0.009 (0.030)	0.192 (0.137)	
Fixed Effects	Y-Mo	Y-Mo	Y-Mo	Y-Mo	S-Y	S-Y	S-Y	S-Y	S-Y

Notes: Y-Mo: Year-month fixed effects included. S-Y: state-year fixed effects included. Mothers and fathers with kid under 5 defined as fathers or mothers with at least one own child age less than 5 present in the household. Table presents estimates of δ from Equation 1 for indicated dependent variables and samples. Robust standard errors in parentheses. Data from 2000-2019 waves of the Current Population Survey, restricted to employed household heads and spouses of household heads age 20-45. Part-time work defined as working positive but fewer than 35 hours in previous week. Snow measured in inches of accumulation in relevant county-month. Illness measured in deaths per 100,000 residents due to flu or pneumonia-related reasons in relevant county-month. See text for details.

Table A.III: Interactions with Number of Children of Various Ages

Panel A: Interactions with Inches of Snow		
VARIABLES	(1) Hours Worked	(2) Part-Time for Childcare
Inches of Snow	0.003 (0.008)	0.014 (0.011)
Snow X # Children 0-5	-0.028 (0.005)	0.009 (0.008)
Snow X # Children 6-11	-0.006 (0.004)	-0.009 (0.007)
Snow X # Children 12-18	-0.004 (0.004)	-0.005 (0.006)
Panel B: Interactions with Flu Intensity		
VARIABLES	(1) Hours Worked	(2) Part-Time for Childcare
Flu Deaths per Capita	-0.020 (0.061)	0.023 (0.093)
Flu X # Children 0-5	-0.042 (0.039)	0.085 (0.068)
Flu X # Children 6-11	-0.008 (0.032)	0.081 (0.052)
Flu X # Children 12-18	0.021 (0.033)	-0.045 (0.048)

Notes: Table presents estimates of impacts of snowfall or flu intensity on indicated dependent variables of interest as well as interactions with number of own children in household age 0-5, 6-11, and 12-18. Data for mothers in 2000-2019 waves of the Current Population Survey, restricted to employed household heads and spouses of household heads age 20-45. Part-time work defined as working positive but fewer than 35 hours in previous week. Snow measured in inches of accumulation in relevant county-month. Illness measured in deaths per 100,000 residents due to flu or pneumonia-related reasons in relevant county-month. Additional controls include year, month, and county fixed effects; Black, white, married, high-school, and college indicators; un-interacted numbers of children; and a cubic polynomial in age. See text for details.

Table A.IV: Summary Statistics for American Time Use Sample

	Mothers	Fathers
Age	38.4	40.6
Married	0.70	0.92
College	0.45	0.46
Black	0.12	0.06
White	0.81	0.86
Childcare minutes	316.7	216.1
Work minutes	360.0	458.3
N	11876	10915

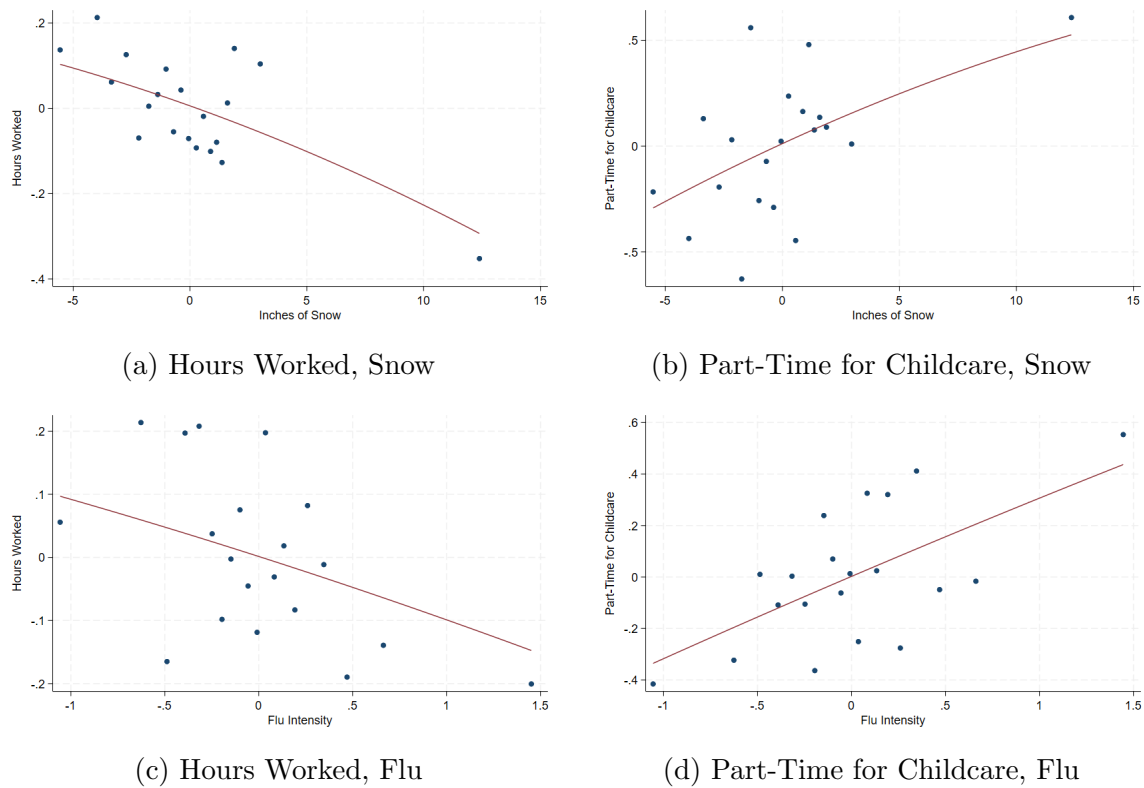
Notes: Table presents means for each variable above for the time use sample. Data is from the 2003-2019 waves of the American Time Use Survey. Childcare minutes include all time spent on both primary childcare (activity category 3) and secondary childcare (*SCC_ALL*); work minutes include all work-related activities (activity category 5); see text for details. Sample is restricted to employed individuals with own child in household.

Table A.V: Downstream Labor Supply Impacts – ACS Sample

Panel A: Effects of Increased Snow							
SAMPLE	All	Men, No Kids	Women, No Kids	Fathers, All	Mothers, All	Fathers, Kid < 5yo	Mothers, Kid < 5yo
VARIABLE							
Wage Income	11.4 (11.2)	11.3 (18.7)	−23.6 (17.8)	55.1 (30.9)	7.69 (19.5)	88.0 (42.9)	−6.31 (27.8)
Employed	−0.013 (0.017)	−0.020 (0.030)	−0.013 (0.035)	0.000 (0.026)	−0.026 (0.034)	−0.003 (0.037)	−0.020 (0.052)
Not in Labor Force	0.002 (0.014)	−0.010 (0.027)	0.041 (0.032)	−0.006 (0.019)	−0.001 (0.032)	−0.008 (0.028)	0.021 (0.050)
Usual Hours Worked	−0.003 (0.007)	−0.014 (0.013)	−0.004 (0.013)	0.009 (0.012)	−0.008 (0.013)	0.014 (0.017)	−0.018 (0.019)
Panel B: Effects of Increased Illness							
VARIABLE							
Wage Income	−61.7 (24.1)	−66.3 (40.9)	−77.3 (38.3)	−116.0 (66.1)	−39.9 (40.7)	−47.7 (92.3)	−13.7 (60.8)
Employed	−0.009 (0.037)	0.079 (0.070)	−0.087 (0.081)	0.002 (0.056)	−0.039 (0.072)	0.065 (0.081)	0.024 (0.116)
Not in Labor Force	0.004 (0.034)	−0.030 (0.061)	0.079 (0.074)	−0.017 (0.043)	−0.004 (0.068)	0.004 (0.061)	−0.061 (0.112)
Usual Hours Worked	−0.013 (0.014)	0.003 (0.028)	−0.048 (0.029)	0.013 (0.026)	−0.012 (0.028)	0.027 (0.037)	−0.003 (0.045)

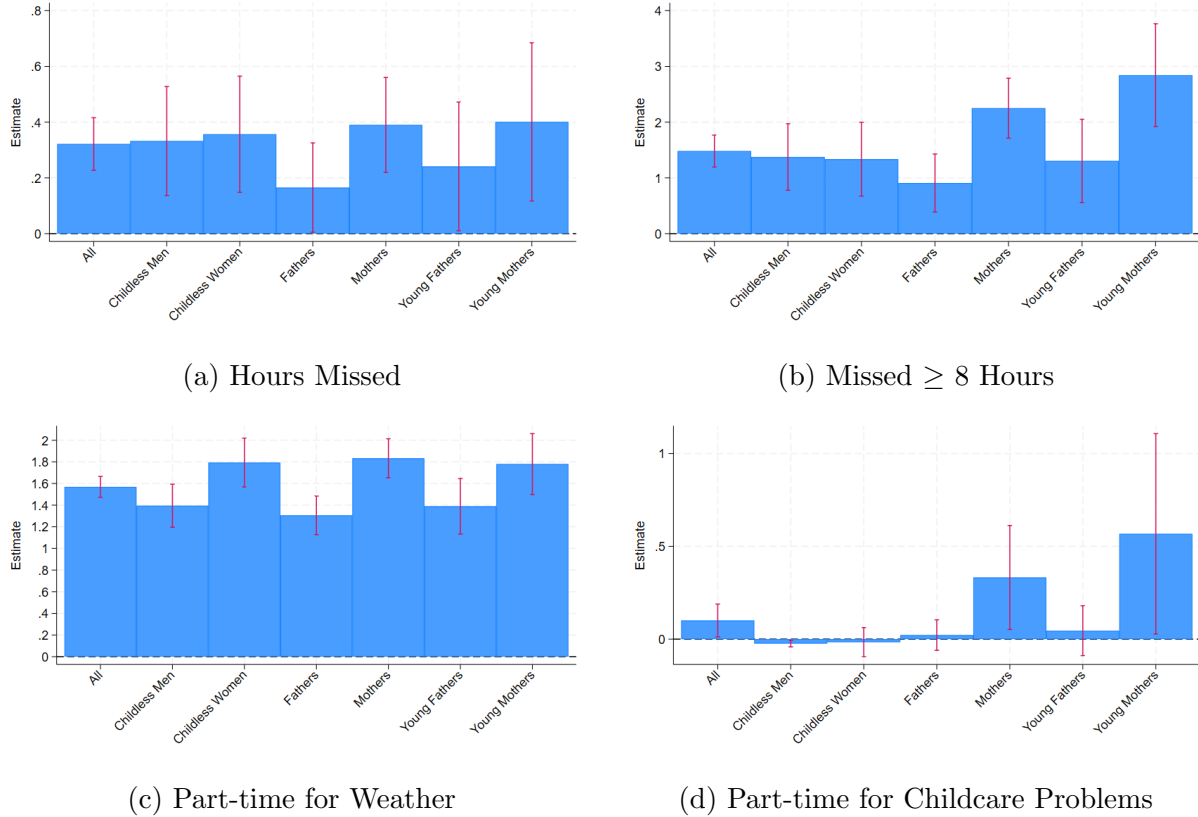
Notes: Table presents estimates of δ from Equation 1 for indicated dependent variables and samples. Robust standard errors in parentheses. Data from 2006-2018 waves of the ACS, restricted to employed household heads and spouses of household heads age 20-45 who did not move across counties in the previous calendar year. Snow measured in feet of accumulation in the September-May of the previous calendar year in the relevant county. Illness measured in average deaths per 100,000 residents due to flu or pneumonia-related reasons in relevant county in the September-May of the previous calendar year. See text for details. Mothers and fathers with kid under 5 defined as fathers or mothers with at least one own child age less than 5 present in the household.

Figure A.I: Binned Scatter Plots of Residualized Dependent and Independent Variables



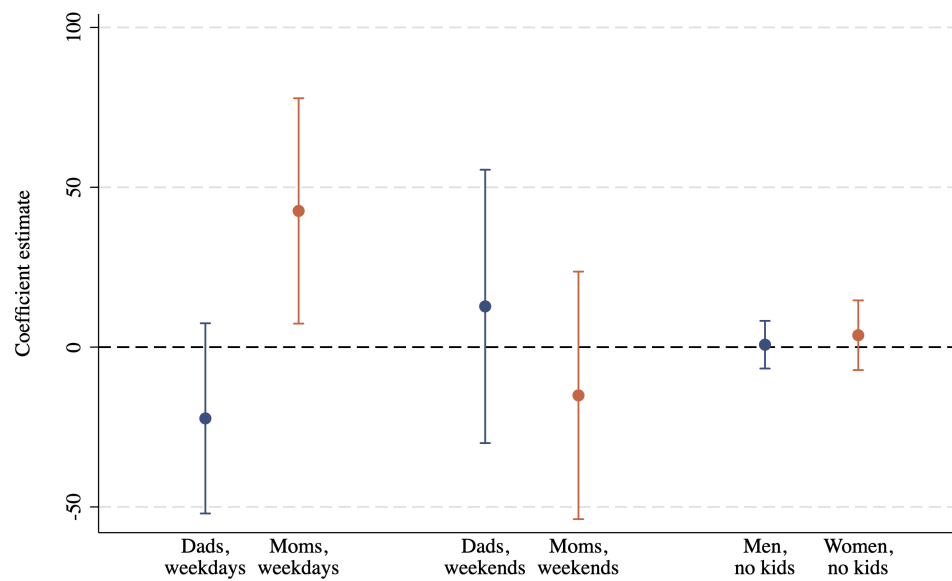
Notes: Figures present binned scatter plots of residualized dependent and independent variables with quadratic lines of best fit. Data for mothers of young children from 2000-2019 waves of the Current Population Survey, restricted to employed household heads and spouses of household heads age 20-45. Part-time work defined as working positive but fewer than 35 hours in previous week. Snow measured in inches of accumulation in relevant county-month. Illness measured in deaths per 100,000 residents due to flu or pneumonia-related reasons in relevant county-month. Variables residualized on year, month, and county fixed effects; Black, white, married, high-school, and college indicators; and a cubic polynomial in age. See text for details.

Figure A.II: Baseline Results — Effects of Increased Snow, 2-SD Measure

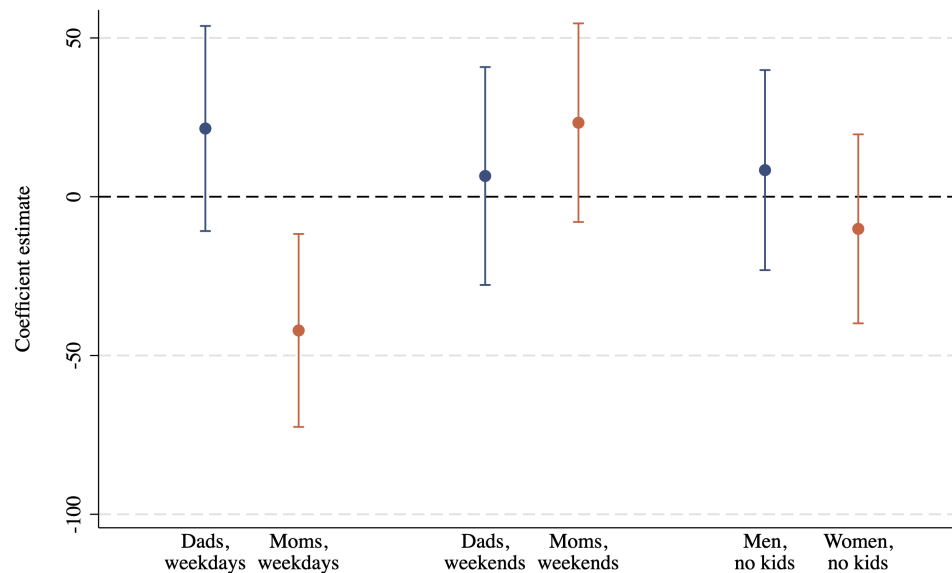


Notes: Young fathers/mothers defined as fathers or mothers with at least one own child age less than 5 present in the household. Figures presents estimates of δ from Equation 1 for indicated dependent variables and samples. 95% confidence intervals in red. Data from 2000-2019 waves of the Current Population Survey, restricted to employed household heads and spouses of household heads age 20-45. Part-time work defined as working positive but fewer than 35 hours in previous week. Treatment is an indicator for a within-county two-standard deviation snowfall. See text for details.

Figure A.III: Validation Comparisons for Parental Time Use Changes



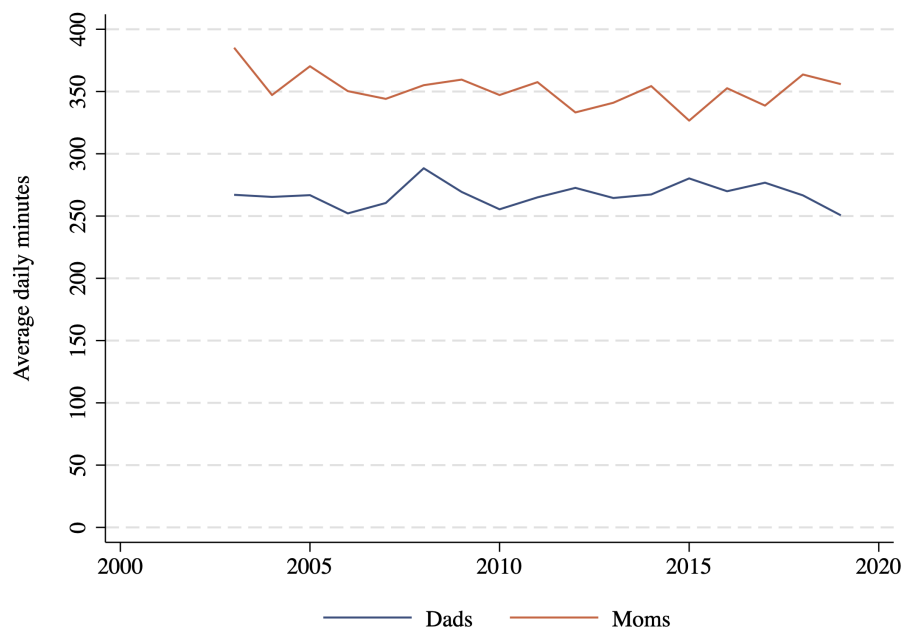
(a) Time Use, Childcare



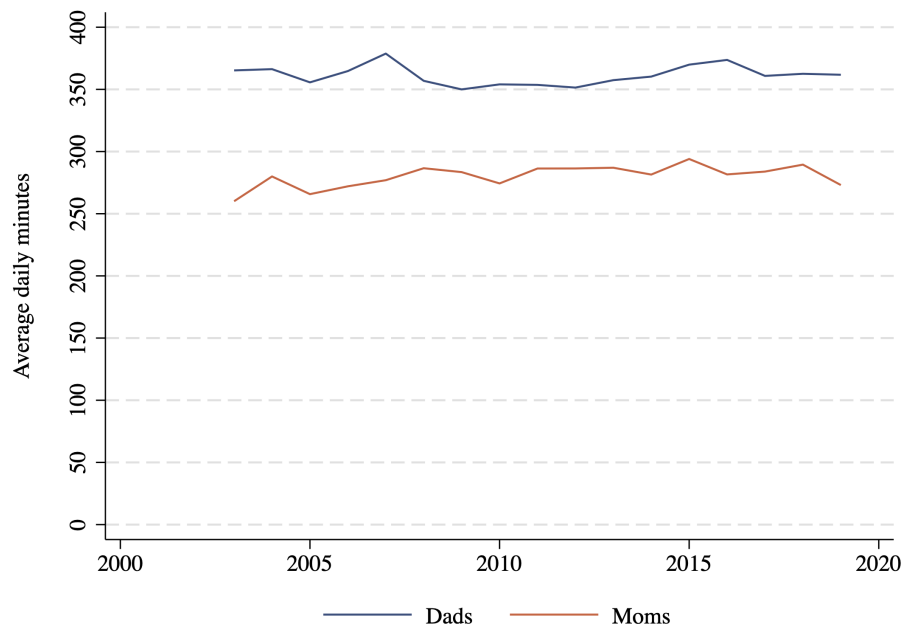
(b) Time Use, Work

Notes: Figures present validation comparisons for the estimated change in time spent on childcare and work presented in Figure 4. Comparison groups are parents on weekends and non-parents. Estimates and confidence intervals are measured as the response to a within-county 2 standard deviation increase in monthly snowfall. Data from 2003-2019 waves of the American Time Use Survey, restricted to employed individuals. Childcare minutes include all time spent on both primary childcare (activity category 3) and secondary childcare (*SCC_ALL*); work minutes include all work-related activities (activity category 5); see text for details.

Figure A.IV: Trends in Parental Time Use, 2003-2019



(a) Time Use, Childcare



(b) Time Use, Work

Notes: Figures present trends in parental time spent on childcare and work by gender from 2003-2019. Data from 2003-2019 waves of the American Time Use Survey, restricted to employed individuals with own child in household. Childcare minutes include all time spent on both primary childcare (activity category 3) and secondary childcare (*SCC-ALL*); work minutes include all work-related activities (activity category 5); see text for details.